Supplementary Materials for "OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability"

Liang Wang
School of Artificial Intelligence and Automation
Huazhong University of Science and Technology

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Contents

1 Supplementary Tables

1.1 Supplementary Table S1: Full Per-Layer JS Divergence

Table 1: Layer-by-layer Jensen-Shannon divergence across training stages. Values are mean pairwise JS across all task pairs.

Layer	Baseline	CPT	v3 (CPT+SFT)	Δ CPT
0	0.160	0.164	0.164	+0.004
1	0.136	0.113	0.113	-0.023
2	0.172	0.164	0.164	-0.008
3	0.151	0.134	0.134	-0.017
4	0.119	0.113	0.113	-0.006
5	0.159	0.160	0.160	+0.001
6	0.131	0.111	0.111	-0.020
7	0.129	0.123	0.123	-0.006
8	0.134	0.126	0.126	-0.008
9	0.174	0.153	0.153	-0.021
10	0.197	0.159	0.159	-0.038
11	0.160	0.143	0.143	-0.017
12	0.182	0.149	0.149	-0.033
13	0.143	0.127	0.127	-0.016
14	0.157	0.137	0.137	-0.020
15	0.198	0.170	0.170	-0.028
16	0.139	0.121	0.121	-0.018
17	0.184	0.166	0.166	-0.018
18	0.148	0.134	0.134	-0.014
19	0.124	0.123	0.123	-0.001
20	0.140	0.130	0.130	-0.010
21	0.136	0.119	0.119	-0.017
22	0.162	0.149	0.149	-0.013
23	0.157	0.138	0.138	-0.019
24	0.141	0.122	0.122	-0.019
25	0.180	0.149	0.149	-0.031
26	0.161	0.152	0.152	-0.009
27	0.178	0.172	0.172	-0.006
28	0.217	0.197	0.197	-0.020
29	0.240	0.222	0.222	-0.018

1.2 Supplementary Table S2: Bootstrap Confidence Intervals

Table 2: Bootstrap 95% confidence intervals for routing metrics (1,000 iterations).

Metric	Mean	95% CI Lower	95% CI Upper
Baseline JS	0.344	0.344	0.344
CPT JS	0.440	0.440	0.441
v3 JS	0.444	0.444	0.444
Δ CPT	0.096	0.096	0.097
Δ SFT	0.004	0.003	0.004

1.3 Supplementary Table S3: Format-Matched Routing Control

Table 3: Layer-averaged JS divergence before and after format matching.

Condition	JS Divergence	Retention Ratio
Original (heterogeneous formats)	0.160	100%
Format-matched (neutral template)	0.145	90.2%

1.4 Supplementary Table S4: ESM-2 Head-to-Head Comparison

Table 4: Remote homology accuracy on identical 2,000-pair evaluation.

Model	Threshold	Accuracy	ROC AUC
ESM-2 (650M)	0.5	50.50%	0.513
ESM-2 (650M)	1.001 (optimal)	51.80%	0.513
OmniGene-4 v3	N/A	59.50%	N/A
Gemma-4 Instruct baseline	N/A	60.00%	N/A

2 Supplementary Figures

2.1 Supplementary Figure S1: Per-Layer JS Curves

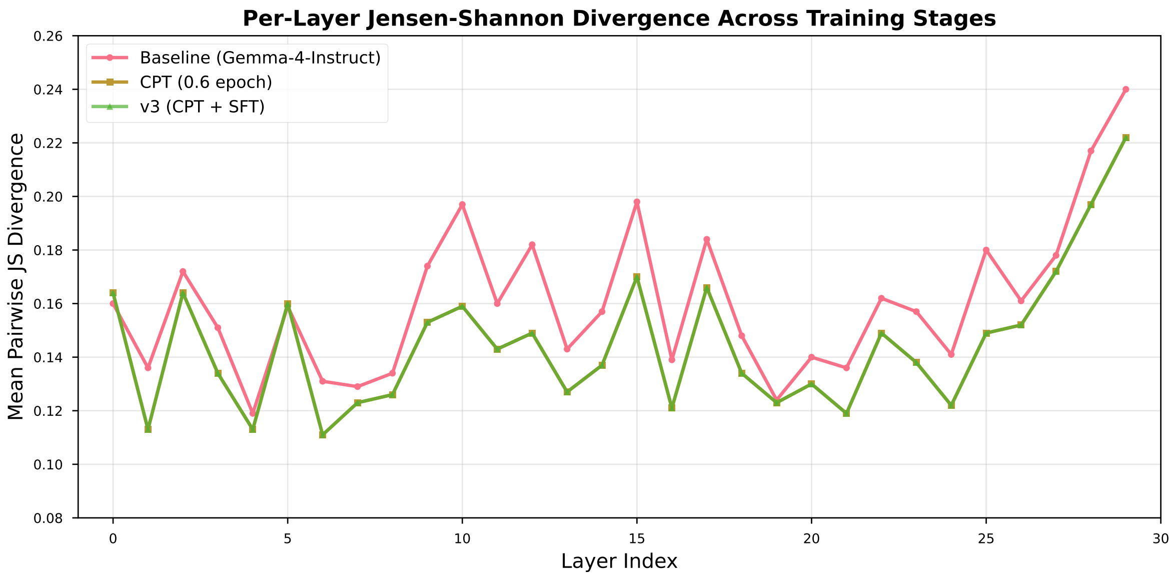

Figure 1: Layer-by-layer Jensen-Shannon divergence across training stages. The CPT stage shows the largest increase in middle layers (L11–L22), while the SFT stage shows concentrated changes near the output layers (L28–L29).

2.2 Supplementary Figure S2: Expert Activation Heatmap

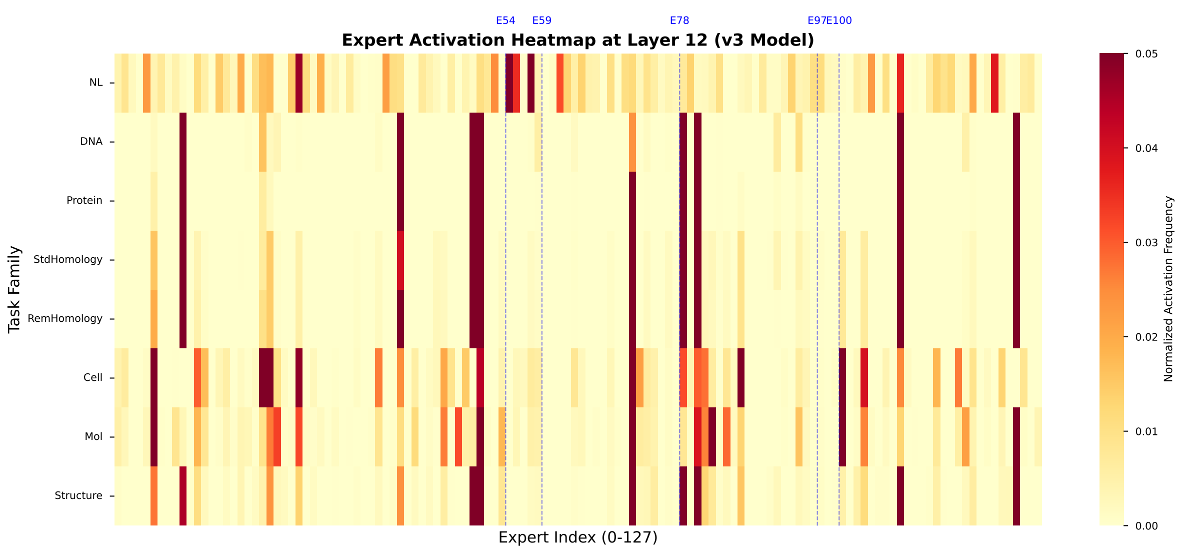

Figure 2: Expert activation heatmap at Layer 12 for eight task families. Rows: tasks (NL, DNA, Protein, StdHom, RemHom, Cell, Mol, Structure). Columns: 128 experts. Color intensity indicates normalized activation frequency. Blue dashed lines mark key experts (E54: English function words, E59: DNA 2-mers, E78: amino acids, E97: cell markers, E100: SMILES punctuation).

2.3 Supplementary Figure S3: Bootstrap Distribution

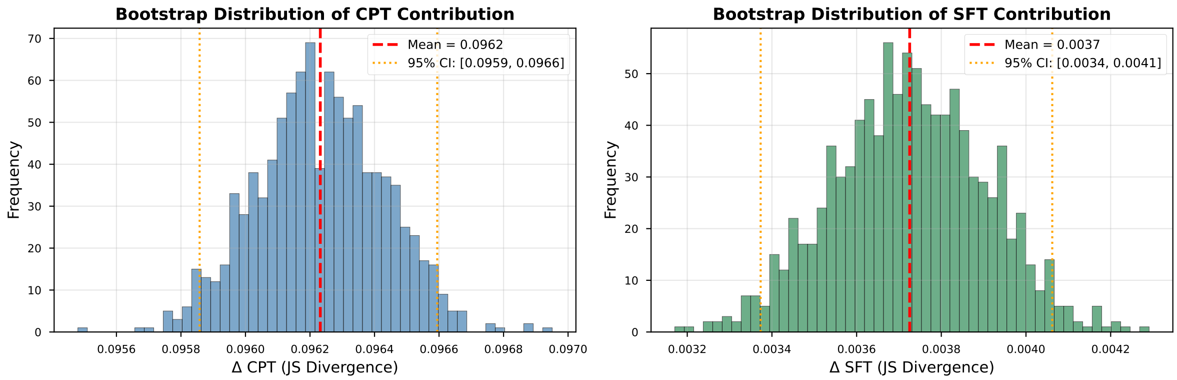

Figure 3: Bootstrap distribution of CPT and SFT contributions to JS divergence (1,000 iterations). Both distributions are narrow and exclude zero, confirming statistical robustness. Left: Δ CPT = 0.096 [0.096, 0.097]. Right: Δ SFT = 0.004 [0.003, 0.004].

3 Supplementary Methods

3.1 Detailed Training Hyperparameters

3.1.1 Continued Pretraining (CPT)

Table 5: CPT hyperparameters for OmniGene-4 v3.

Parameter	**Value**
Base model	Gemma-4-26B-A4B-Instruct
Vocabulary size	290,048 (262,020 + 28,028 bio tokens)
Training data	32.5 GB (8.73B tokens)
Batch size	6 per GPU $\times$ 8 GPUs $\times$ 4 accum = 192
Learning rate	2e-5
LR scheduler	Cosine with 3% warmup
Optimizer	paged_adamw_8bit
Weight decay	0.01
Max gradient norm	1.0
Sequence length	1024
Epochs	0.6
Total steps	2,806
LoRA rank	64
LoRA alpha	128
LoRA dropout	0.05
LoRA targets	q/k/v/o_proj, gate/up/down_proj, router.proj
Precision	bfloat16
Gradient checkpointing	True

3.1.2 Supervised Fine-Tuning (SFT)

Table 6: SFT hyperparameters for OmniGene-4 v3.

Parameter	Value
Training data	199,576 instruction examples
Batch size	4 per GPU × 1 GPU × 16 accum = 64
Learning rate	5e-5
LR scheduler	Cosine with 3% warmup
Optimizer	paged_adamw_8bit
Weight decay	0.01
Max gradient norm	1.0
Sequence length	1024
Epochs	1
Total steps	3,118
LoRA rank	64
LoRA alpha	128
LoRA dropout	0.05
LoRA targets	q/k/v/o_proj, gate/up/down_proj, router.proj
Precision	bfloat16
Gradient checkpointing	True

3.2 Data Mixture Composition

3.2.1 CPT Data Mixture

Table 7: CPT data mixture composition (32.5 GB total).

Source	Size (GB)	Tokens (B)	Proportion
DNA (human genome)	8.0	2.1	24.6%
Protein (UniProt)	8.0	2.1	24.6%
Protein (LucaOne)	7.5	2.0	23.1%
OpenWebText	8.0	2.1	24.6%
Structure (3Di + DSSP)	0.4	0.1	1.2%
Instruction replay	0.6	0.4	1.9%
Total	**32.5**	**8.7**	**100%**

3.2.2 SFT Data Mixture

Table 8: SFT data mixture composition (199,576 examples).

Task Family	Examples	Proportion
Protein homology	49,894	25.0%
Literature (UniProtQA)	39,915	20.0%
Mutation (MutaDescribe)	29,936	15.0%
Cell biology	29,936	15.0%
Molecule (SMILES)	25,945	13.0%
Structure (3D)	19,958	10.0%
DNA homology	3,992	2.0%
Total	**199,576**	**100%**

3.3 Routing Collection Protocol

3.3.1 Forward Hook Implementation

Listing 1: Forward hook for collecting router activations

```
def make_hook(layer_idx):
    def hook(module, input, output):
        # output = (router_probs, top_k_weights, top_k_indices)
        # top_k_indices shape: [seq_len, 8]
        top_k_indices = output[2]
        for seq_idx in range(top_k_indices.shape[0]):
            for k_idx in range(top_k_indices.shape[1]):
                expert_id = int(top_k_indices[seq_idx, k_idx].item())
                routing_counts[current_task][layer_idx, expert_id] += 1
    return hook

# Register hooks on all layers
handles = []
for i in range(NUM_LAYERS):
    h = model.model.language_model.layers[i].router.register_forward_hook(
        make_hook(i))
    handles.append(h)
```

3.3.2 Bootstrap Resampling Protocol

Listing 2: Prompt-level bootstrap resampling

```
def bootstrap_sample(prompt_counts_dict, tasks):
    """
    prompt_counts_dict: {task: [n_prompts, layers, experts]}
    Returns: resampled counts_dict {task: [layers, experts]}
    """
    resampled = {}
    for task in tasks:
        prompts = prompt_counts_dict[task]
```

```python
        n_prompts = prompts.shape[0]
        # Resample with replacement
        indices = np.random.choice(n_prompts, n_prompts, replace=True)
        resampled_prompts = prompts[indices]
        # Sum over prompts
        resampled[task] = resampled_prompts.sum(axis=0)
    return resampled

# Run 1000 bootstrap iterations
for i in range(1000):
    baseline_boot = bootstrap_sample(baseline_prompts, tasks)
    cpt_boot = bootstrap_sample(cpt_prompts, tasks)
    v3_boot = bootstrap_sample(v3_prompts, tasks)
    # Compute JS and store
    ...
```

4 Supplementary Results

4.1 Token-Level Expert Specialization

Table 9: Top-5 tokens for selected experts at Layer 12 (v3 model).

Expert	Purity	Top-5 Tokens
E54	80%	`the`, `of`, `and`, `to`, `in`
E59	46%	`AA`, `AT`, `TT`, `GG`, `CC`
E78	35%	`M`, `L`, `V`, `I`, `F`
E97	28%	`CD4`, `CD8`, `IL`, `TNF`, `IFN`
E100	15%	`(`, `)`, `=`, `C`, `O`

4.2 Per-Task Entropy Analysis

Table 10: Shannon entropy of expert distributions across training stages.

Task	Baseline	CPT	v3 (CPT+SFT)
NL	3.85	4.13	4.15
DNA	2.89	2.76	2.78
Protein	2.76	2.54	2.56
StdHomology	2.68	2.48	2.50
RemHomology	2.71	2.52	2.54
Cell	2.95	2.88	2.90
Mol	3.02	2.95	2.97
Structure	2.82	2.65	2.67

5 Code Availability

All code, training scripts, and analysis notebooks are available at:

`https://github.com/maris205/omnigene4`

5.1 Key Scripts

- `biopaws/cpt/1-prepare_cpt_data_mp.py` – CPT data preparation (multi-process)
- `biopaws/cpt/2-run_cpt.py` – CPT training (8-GPU DDP)
- `biopaws/sft_data/17-train_bio_sft_v3_remote.py` – SFT training
- `biopaws/cpt/18-eval_v3_sft.py` – Main evaluation
- `biopaws/cpt/20-collect_moe_activations.py` – Routing collection
- `biopaws/cpt/26-format_matched_routing.py` – Format control
- `biopaws/cpt/27-esm2_remote_headtohead.py` – ESM-2 comparison
- `biopaws/cpt/28-bootstrap_ci.py` – Bootstrap CI

5.2 Data Files

Routing-count matrices and analysis results are in `outputs/moe_analysis/`:

- `moe_counts_baseline.npz` – Baseline routing (50 KB)
- `moe_counts_cpt.npz` – CPT routing (47 KB)
- `moe_counts_v3.npz` – v3 routing (48 KB)
- `format_matched_report.json` – Format control results
- `esm2_remote_2k_eval.json` – ESM-2 comparison results
- `bootstrap_ci_report.json` – Bootstrap CI results
- `bootstrap_samples.npz` – Full bootstrap samples (1,000 iterations)

5.3 Model Weights

LoRA adapter weights and embedding checkpoints ($\approx$ 1.9 GB) can be released upon reasonable request to the corresponding author.